

Vision Documentation

Interactive House - AI Smart Control and Safety System

Revision History

Date	Version	Description	Author
2026-02-24	1.0	Initial AI-based vision documentation (1-2)	Md Abu Taher
2026-02-24	1.0	Initial AI-based vision documentation (3-4)	Anitta Antony
2026-02-28	1.0	Smart Home code	Daniel Marcarini
2026-02-28	1.0	Speech Recognition Implementation	Daniel Holderik
2026-03-16	1.1	Updated Vision with Firebase AI and Web	
2026-03-16	1.1	Testing	Daniel Marcarini
2026-04-22	1.2	Updated with emergency button	Anitta Antony Md Abu Taher
2026-05-08	1.3	Final update to align vision with AI control, web/mobile interaction, emergency assistance	Anitta Antony Md Abu Taher
2026-05-08	1.4	Updated vision with current automated web testing status using Jest and Istanbul coverage	Daniel Marcarini

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Product Overview

1. System Concept

1.1 Stakeholders

The primary stakeholders of the Interactive House system include:

- The development team responsible for system design and implementation.
- Course supervisors and examiners evaluating the innovation and software engineering quality.
- End users, represented by students and individuals interested in smart home technologies.
- Future homeowners who may benefit from intelligent home automation and safety monitoring solutions.

For development and testing purposes, student peers act as representative end users to simulate real-world interaction scenarios.

1.2 Introduction

The Interactive House system is an AI-driven smart home solution designed to integrate

natural language interaction with automated device control and safety monitoring.

The core idea is to allow users to control home devices such as doors and ventilation systems using simple conversational commands via a web or mobile application. In addition to manual interaction, the system continuously monitors environmental and safety-related conditions, including fire detection, gas leakage, rain detection, and indoor temperature.

By combining artificial intelligence with sensor-based automation, the system aims to enhance safety, comfort, and usability within a simulated smart home environment.

The project emphasizes innovation through AI integration while maintaining structured software engineering principles.

2. System Rules and Behavior

The system operates according to predefined control and safety logic.

The primary behavioral rules are:

- The system interprets user commands through AI processing and forwards structured intent to the device control layer for execution.
- The system design prioritizes safety over user-initiated actions in cases of conflict. Conflict resolution logic is handled at the device/safety layer.
- Device states shall be maintained and updated consistently.
- Environmental data such as temperature and rain detection shall influence system recommendations or automated responses.

The system prioritizes safety over user-initiated actions when conflicts occur.

3. System Technical Structure

The Interactive House system integrates multiple technical components to achieve intelligent automation.

3.1 General Structure

The system consists of:

- A web and mobile application interface for user interaction.
- A Firebase based AI service for natural language interpretation.
- Simulated smart devices and sensors.

The architecture follows modular design principles to ensure maintainability and scalability.

3.2 Normal Operation Mode

In normal operation mode, users communicate with the system through simple natural language commands using the AI chat interface in the web or mobile application. users communicate with the system through simple natural language commands using the mobile AI interface, including both text and speech input.

This mode allows users to request actions related to devices such as doors and ventilation, as well as ask about temperature or general system status. The application displays the AI's response as confirmation or information for the user.

3.3 Safety Monitoring Mode

The system is designed to monitor safety-related sensors such as fire detection and gas leakage. When an abnormal condition is detected, the appropriate alarm can be activated automatically and the user can be notified.

In certain situations, additional protective actions may be triggered, such as activating ventilation in the case of gas detection. This safety functionality is architecturally separated from the AI chat interface to ensure that emergency responses do not depend on user commands.

3.4 AI Interaction Mode

The AI interaction mode allows users to communicate with the system using natural language through the chat interface in the web or mobile application. The AI interprets different ways of phrasing commands and generates appropriate responses.

If a command is unclear, the AI provides feedback or clarification in conversational form. The current implementation processes each request independently, while the design allows future extensions to support more advanced contextual interactions and additional smart devices.

4. Basic Requirements

The following represent the core expectations from stakeholders:

- The system shall support natural language command processing through the AI interface.
- The system shall enable reliable device control through the AI service and device control layer.
- The system is designed to support automatic handling of safety-critical events through dedicated sensor logic.
- The system shall provide clear and meaningful feedback to users through the application interface.
- The system architecture shall support future expansion and integration of additional devices and AI capabilities.
- The system shall provide an emergency assistance feature that allows users to quickly activate a simulated emergency call or alert flow. This feature is intended to support contact with a caretaker or responsible person in an emergency scenario, without integrating with real emergency services.

The final scope of the SG4 Innovation contribution includes AI-based interaction, web and mobile user interfaces, Firebase AI integration, device-control support, speech interaction where platform support allows it, and an emergency assistance simulation. The project remains a

prototype for academic demonstration, so real emergency-service integration and full production-level smart-home deployment are outside the current scope.

The system shall support structured testing of important web application flows so that authentication, Firebase configuration, and later device-control behavior can be verified before demonstration.

Relation with Other Artefacts

The vision document defines the overall goal and scope of the Interactive House system. The requirements document breaks this vision into functional requirements such as AI command processing, device control, speech support, safety monitoring, and emergency assistance. The design document explains how these features are structured through the AI interface, Firebase AI integration, device-state handling, speech recognition, and emergency flow. The risk document identifies possible problems related to AI interpretation, response delay, external service dependency, and speech/browser limitations. The supplementary requirements define quality expectations such as usability, reliability, performance, security, maintainability, and testability.

The current project work also includes an initial automated testing setup for the web application. Jest and React Testing Library are used to verify important authentication flows and Firebase configuration behavior in a repeatable way. Istanbul coverage reporting is used to display the current automated test coverage. This supports the project vision by improving confidence in the reliability of the web interface and by providing a foundation for future automated testing of device control, emergency, music, and AI-related features.

Daniel Marcarini - Smart Home code

Vision Document Content – Smart House System

1. Introduction

The purpose of this project is to design and implement a Smart House prototype using an Arduino-based embedded system. The system aims to simulate intelligent home functionality by integrating sensors, actuators, and user interaction components into a unified control system.

The project focuses on demonstrating core smart-home concepts such as environmental monitoring, automated responses, and user-controlled features in a simplified but functional prototype.

2. Product Positioning

The Smart House system is a prototype designed for educational and demonstrative purposes. It is not intended to replace commercial home automation systems but to illustrate how embedded systems can enhance safety, comfort, and interaction within a residential environment.

The system integrates:

- Gas detection and safety response
- Temperature monitoring
- Light detection
- Servo-controlled mechanical elements (door and window)
- LCD feedback display
- Audible alarm system
- Bluetooth communication for remote interaction

3. Stakeholders

The primary stakeholders include:

- End users who interact with the Smart House system.
- Developers responsible for implementing and maintaining the embedded system.
- Academic evaluators assessing system functionality and documentation quality.

The system is designed with usability and clarity in mind, ensuring that interaction through buttons, sensors, and display feedback is intuitive and understandable.

4. Product Overview

The Smart House prototype consists of the following major components:

- **Input Components**
 - Gas sensor
 - Temperature sensor
 - Light-dependent resistor (LDR)
 - Push buttons
 - Bluetooth module
- **Processing Unit**
 - Arduino Uno microcontroller
- **Output Components**
 - Servo motors (door and window)
 - LCD display
 - Buzzer (multiple sound modes)
 - LEDs

The system continuously monitors environmental conditions and reacts according to predefined behavioral logic.

5. Key Features

Safety Monitoring

The system detects high gas levels and activates a staged alarm procedure. This includes:

- Immediate audible warning
- Visual indication
- Automatic mechanical response (e.g., opening window)
- Controlled timing behavior using non-blocking logic

Environmental Awareness

Temperature and light levels are monitored and displayed. The system provides visual feedback via LCD and LED indicators.

User Interaction

Users can:

- Control the window and door using push buttons
- Interact remotely via Bluetooth
- Receive real-time system status on the LCD display

Alert System

The buzzer supports multiple operational modes:

- Continuous alarm
- Siren-style oscillating sound
- Silent mode

A centralized buzzer control mechanism ensures that sound behaviors do not conflict.

6. Business Opportunity / Educational Value

Although this system is developed as an academic prototype, it demonstrates principles used in real smart-home systems:

- Event-driven programming
- Embedded system architecture
- State-based control logic
- Non-blocking timing using millis()
- Hardware-software integration

The project highlights how microcontroller-based systems can manage multiple subsystems simultaneously in a reliable and structured way.

7. Product Scope

The system includes:

- Real-time monitoring of environmental sensors
- Automated safety response to gas detection
- Manual and remote control of actuators
- Visual and auditory feedback
- Structured system state management

The system does not include:

- Internet/cloud integration
- Mobile application interface
- Persistent data storage
- Advanced AI-based automation

The scope is intentionally limited to ensure reliability and clarity of core functionality within an embedded system context.